

Writing Recursive Rules for Arithmetic Sequences

Answer Key

Algebra 1

Quick Answers

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|---|---|
| 1. the step $d = 7$, the first term = 4, the tenth term = 67 | 6. saved per week = 20, balance after week 1 = 105, balance after week 12 = 325 |
| 2. the step $d = -4$, the first term = 28, the ninth term = -4 | 7. the tenth term = 59, sum of the first ten terms = 320 |
| 3. seats added per row = 3, seats in row 1 = 9, seats in row 12 = 42 | 8. the step $d = 4$, the first term = -5 , the eleventh term = 35 |
| 4. second term = 11, third term = 16, fourth term = 21, twentieth term = 101, — | 9. rise per rung = 15, height of rung 1 = 19, height of rung 10 = 154 cm |
| 5. the step $d = 5$, the first term = -17 , the fifteenth term = 53 | 10. the term number = 7, the common value = 33 |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 4. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSF-BF.A.2 – *problems 1, 2, 4, 5, 7, 8, 10*

HSF-LE.A.2 – *problems 3, 6, 9*

Difficulty 1–5 of 5 across 30 tagged checks.

Common wrong answers

- If they got -7 : subtracted the later term from the earlier one*
 - If they got 6: counted four steps back from row 4 instead of three*
 - If they got 39: multiplied the step by the term number instead of one less*
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- Second, third and fourth terms 11, 18, 25. Find d , a_1 , and a_{10} .

Solution: Consecutive terms differ by the step, so $18 - 11$ gives it directly.

Term 2 is one step past term 1, so walk back: $11 - 7$.

$d = 7$

$a_1 = 4$

The recursive rule is $a_1 = 4$, $a_n = a_{n-1} + 7$ for $n \geq 2$. Reaching term 10 takes nine steps from term 1: $4 + 9(7) = 4 + 63$.

$$a_{10} = 67$$

2. $a_3 = 20$ and $a_6 = 8$. Find d , a_1 , and a_9 .

Solution: Terms 3 and 6 are three steps apart, so the step is $\frac{8-20}{3} = \frac{-12}{3}$.

$$d = -4$$

Walk back two steps from term 3: $20 - 2(-4) = 20 + 8$.

$$a_1 = 28$$

The rule is $a_1 = 28$, $a_n = a_{n-1} - 4$. Term 9 is eight steps on: $28 + 8(-4) = 28 - 32$.

$$a_9 = -4$$

3. Theatre rows: row 4 holds 18 seats, row 8 holds 30. Find d , row 1, and row 12.

Solution: Rows 4 and 8 are four rows apart: $\frac{30-18}{4} = \frac{12}{4}$.

$$d = 3$$

Row 1 is three rows back from row 4: $18 - 3(3) = 18 - 9$.

$$a_1 = 9$$

With $a_1 = 9$ and $a_n = a_{n-1} + 3$, row 12 is eleven steps on: $9 + 11(3) = 9 + 33$.

$$a_{12} = 42$$

4. $a_1 = 6$, $a_n = a_{n-1} + 5$. (a) Next three terms. (b) a_{20} . (c) Why the step alone is not enough.

Solution (a): Add 5 each time. First step: $6 + 5$.

$$a_2 = 11$$

Second step: $11 + 5$.

$$a_3 = 16$$

Third step: $16 + 5$.

$$a_4 = 21$$

Solution (b): Term 20 is nineteen steps past term 1, so $6 + 19(5) = 6 + 95$.

$$a_{20} = 101$$

(c) — **instructor-judged.** A correct answer says the step rule alone leaves the sequence undetermined: it says every term is five more than the one before, but without a stated starting value there is one such sequence through every possible first number. The recursive rule needs both an anchor and a step. Full credit for naming the missing starting value and saying that infinitely many sequences fit the step alone; partial credit for saying only that the first term is missing.

5. $a_5 = 3$ and $a_9 = 23$. Find d , a_1 , and a_{15} .

Solution: Terms 5 and 9 are four steps apart: $\frac{23-3}{4} = \frac{20}{4}$.

$$d = 5$$

Walk back four steps from term 5: $3 - 4(5) = 3 - 20$.

$$a_1 = -17$$

With $a_1 = -17$ and $a_n = a_{n-1} + 5$, term 15 is fourteen steps on: $-17 + 14(5) = -17 + 70$.

$$a_{15} = 53$$

6. Weekly deposits: balance \$145 after week 3, \$225 after week 7. Find d , a_1 , and the balance after week 12.

Solution: Weeks 3 and 7 are four deposits apart: $\frac{225-145}{4} = \frac{80}{4}$.

$$d = 20$$

Walk back two weeks from week 3: $145 - 2(20) = 145 - 40$.

$$a_1 = 105$$

With $a_1 = 105$ and $a_n = a_{n-1} + 20$, week 12 is eleven steps on: $105 + 11(20) = 105 + 220$.

$$a_{12} = 325$$

7. $a_1 = 5$, $a_n = a_{n-1} + 6$. (a) Find a_{10} . (b) Find the sum of the first ten terms.

Solution (a): Nine steps past the start: $5 + 9(6) = 5 + 54$.

$$a_{10} = 59$$

Solution (b): Pair the first and last terms: an arithmetic sum of n terms is $\frac{n}{2}(a_1 + a_n)$, here $\frac{10}{2}(5 + 59) = 5(64)$.

$$\text{Sum} = 320$$

8. $a_2 = -1$ and $a_6 = 15$. Find d , a_1 , and a_{11} .

Solution: Terms 2 and 6 are four steps apart: $\frac{15 - (-1)}{4} = \frac{16}{4}$.

$$d = 4$$

Walk back one step from term 2: $-1 - 4$.

$$a_1 = -5$$

With $a_1 = -5$ and $a_n = a_{n-1} + 4$, term 11 is *ten* steps on, not eleven: $-5 + 10(4) = -5 + 40$.

$$a_{11} = 35$$

9. Ladder rungs: rung 2 at 34 cm, rung 5 at 79 cm. Find d , rung 1, and rung 10.

Solution: Rungs 2 and 5 are three gaps apart: $\frac{79 - 34}{3} = \frac{45}{3}$.

$$d = 15$$

Rung 1 is one gap below rung 2: $34 - 15$.

$$a_1 = 19$$

With $a_1 = 19$ and $a_n = a_{n-1} + 15$, rung 10 is nine gaps up: $19 + 9(15) = 19 + 135$.

$$a_{10} = 154 \text{ cm}$$

10. Sequence A : $a_1 = 3$, step $+5$. Sequence B : $b_1 = 57$, step -4 . Where are they equal, and at what value?

Solution (a): Write each term with its number of steps: $a_n = 3 + 5(n - 1)$ and $b_n = 57 - 4(n - 1)$. Setting them equal, $3 + 5(n - 1) = 57 - 4(n - 1)$, so $9(n - 1) = 54$ and $n - 1 = 6$.

$$n = 7$$

Solution (b): Substitute back: $3 + 5(6) = 3 + 30$. Check with B : $57 - 4(6) = 57 - 24 = 33$

✓

$$\text{value} = 33$$